

GOSIM Paris 2026

May 5-6, 2026 Station F, Paris



HiFloat8: A Tapered Format Complementing the FP8 Ecosystem for Robust Model Training

Robust Open-Source LLM & Agent Deployment



CONTENTS

GOSIM Paris 2026

Part 01

Background

Part 02

Introduction of HiF8

Part 03

Technical Implementation

Part 04

Conclusion



01

Background :

The Core Bottleneck





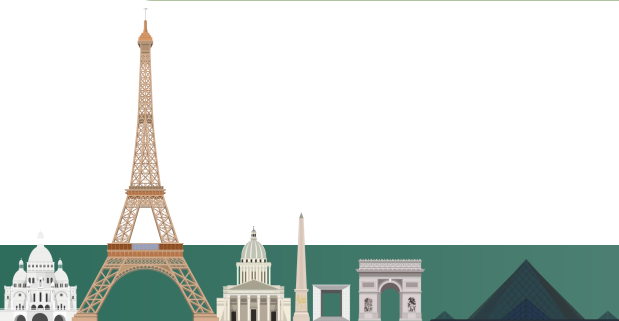
Core Bottleneck

- A core bottleneck for agentic frameworks, edge deployments, and production-grade model services at GOSIM is the VRAM and compute cost of LLM training and inference.
- High computational demands directly hinder scalability and increase operational costs, creating a critical barrier to the widespread adoption of large models in practical, real-world environments.



Industry Status & Talk Preview

- 8-bit quantization is mainstream, but standard FP8 still struggles with instability, complexity, and poor edge adaptability across different hardware.
- In this talk, we will present HiFloat8—an open-source solution solving these pain points at the format level. It is fully compatible with the existing FP8 ecosystem and offers zero-threshold integration with PyTorch.





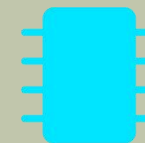
Instability

Frequent gradient overflows. Fixed bit-width design cannot balance the dynamic range of weights/activations and gradients, leading to convergence anomalies in billion-parameter models.



Complexity

Heavy reliance on complex dynamic scaling. Requires per-block scaling strategies for each layer, making hyperparameter tuning costly and incompatible with mature FP16 training paradigms.



Adaptability

High edge/device adaptation difficulty. The dual-format (E4M3/E5M2) design increases hardware and inference framework adaptation costs, hindering agent deployment to edge devices.



02

Introduction of HiF8



HiF8



Core Position

A new 8-bit floating-point format designed for deep learning. It does not replace standard FP8, but provides a robust, easy-to-deploy supplement to the FP8 ecosystem.



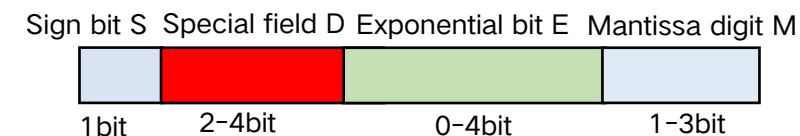
Core Design

An innovative tapered precision encoding architecture that dynamically allocates exponent and mantissa bits to naturally match the clustered distribution of neural network weights, activations, and gradients.



Core Capability

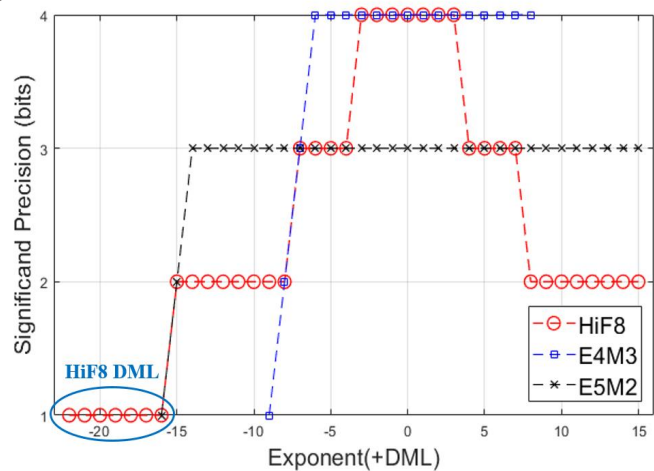
A single format covers **both forward and backward** training passes without switching. It can capture high-magnitude outliers without the aggressive scaling required by standard FP8.



S sign bit sum
The first digit of the
E exponent, with a
positive number of
0.
The negative
number is 1.

D: 11; E: 4 bits; E: x000-x111: $2 \pm 8 - 2 \pm 15$
D: 10; E: 3 bits; E: x00-x11: $2 \pm 4 - 2 \pm 7$
D: 01; E: 2 bits; E: x0-x1 indicates $2 \pm 2 - 2 \pm 3$.
D: 001; E: 1 bit; E: 0-1 indicates 2 ± 1 .
D: 0001; E: 20
D: 0000, indicating the DML range.





The HiF8 optimizes the distribution feature, achieves the cone precision, and matches the AI parameter distribution feature.

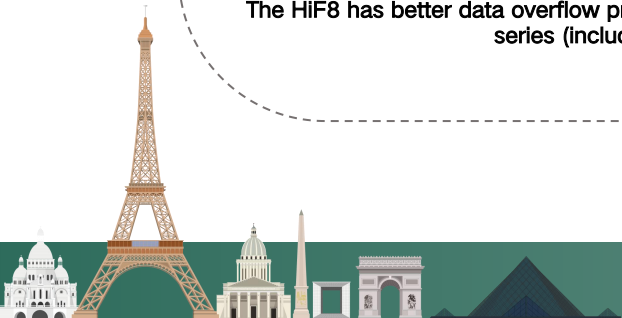
	FP16	HiF8	FP8-E4M3	FP8-E5M2
Infinites	Support	$S1101111_2$	N/A	Support
NaN	Support	1000000_2	Support	Support
Zero	Support	0000000_2	Support	Support
Max Positive NML	$2^{15} \times (2 - 2^{-10})$	$01101110_2 = 2^{15}$	1.75×2^8	1.75×2^{15}
Min Positive NML	2^{-14}	$01111110_2 = 2^{-15}$	2^{-6}	2^{-14}
Max Positive DML	$2^{-14} \times (1 - 2^{-10})$	$00000111_2 = 2^{-16}$	1.75×2^{-7}	1.5×2^{-16}
Min Positive DML	2^{-24}	$00000001_2 = 2^{-22}$	2^{-9}	2^{-16}
Exponent	[-14, 15]	[-15, 15]	[-6, 8]	[-14, 15]
Exponent (+DML)	[-24, 15]	[-22, 15]	[-9, 8]	[-16, 15]

The HiF8 has better data overflow prevention capabilities than the FP8 series (including DML).

Dot	Dot Field	Mode	Exponent	Mantissa	Exponent Field
0	0001	NML	0 bit	3 bit	0
1	001		1 bit (SE)	3 bit	± 1
2	01		2 bit (SE+E)	3 bit	$\pm [2, 3]$
3	10		3 bit (SE+E+E)	2 bit	$\pm [4, 7]$
4	11		4 bit (SE+E+E+E)	1 bit	$\pm [8, 15]$
DML	0000	DML	0 bit	3 bit	$[-22, -16]$

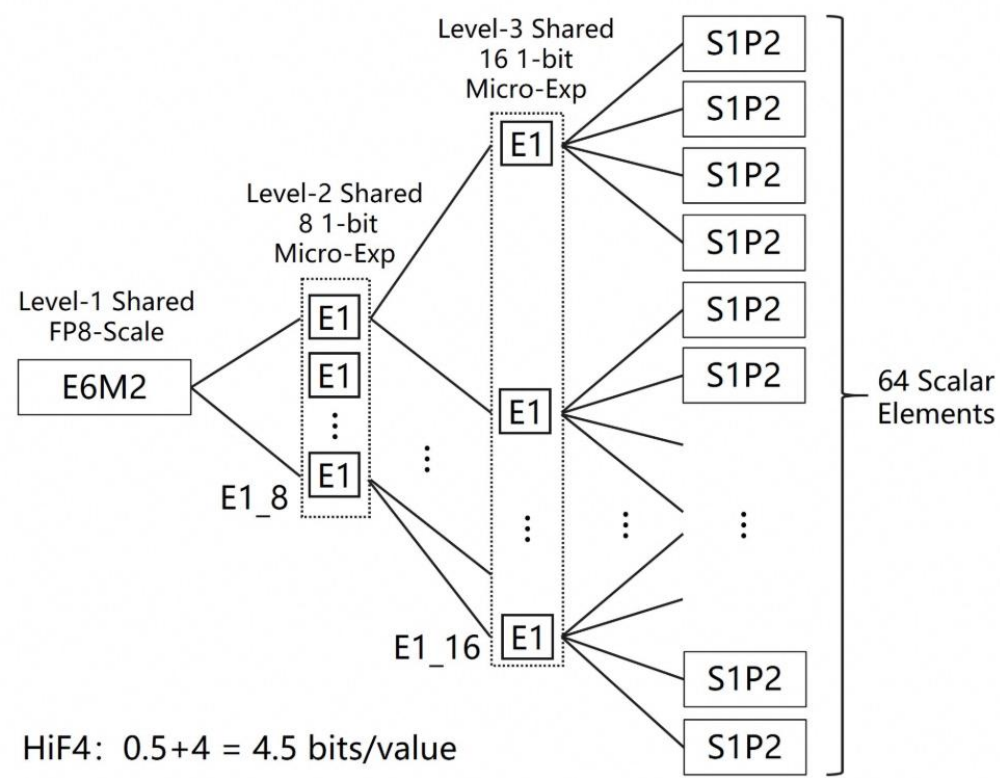
How to use the HiF8:

- The existing FP8 solutions (such as the scaling algorithm and training framework) are compatible with HiF8, including delayed/current scaling at various granularities. In terms of format design advantages, the solution design is more tolerant and the precision-performance advantage is obtained.
- Training: More tolerant of scale frequency requirements and more stable training
- Inference: The calibration difficulty is lower than that of FP8 E4M3, and the same inference precision can be achieved with lower calibration complexity (overhead). or higher inference precision can be obtained under the same calibration overhead.



DIMENSION	HIFLOAT8	STANDARD FP8 (E4M3+E5M2)	FP16
Training/Inference Coverage	Single format full coverage	Requires dual-format switching	Single format full coverage
Dynamic Range	38 binades	Up to 32 binades	40 binades
Scaling Dependency	Low delayed scaling (reuses FP16)	Heavy reliance on per-block scaling	Global loss scaling only
Hardware Adaptation Cost	Single codec logic (simple)	Dual logic & complex scheduling	Single logic (mature)
Memory Footprint	8bit, 50% reduction vs FP16	8bit, 50% reduction vs FP16	16bit baseline





	Unsigned FP8-E6M2	Sign-Magnitude S1P2
Exponent Bias	48	N/A
Unbiased Exp	$[-48, 15]$	N/A
Infinity	N/A	N/A
Zero	N/A	$S0.00_2 = \pm 0.00$
NaN	111111_11_2	N/A
Max Value	$111111_10_2 = 2^{15} \times 1.50$	$S1.11_2 = \pm 1.75$
Min Value	$000000_00_2 = 2^{-48} \times 1.00$	$S0.01_2 = \pm 0.25$

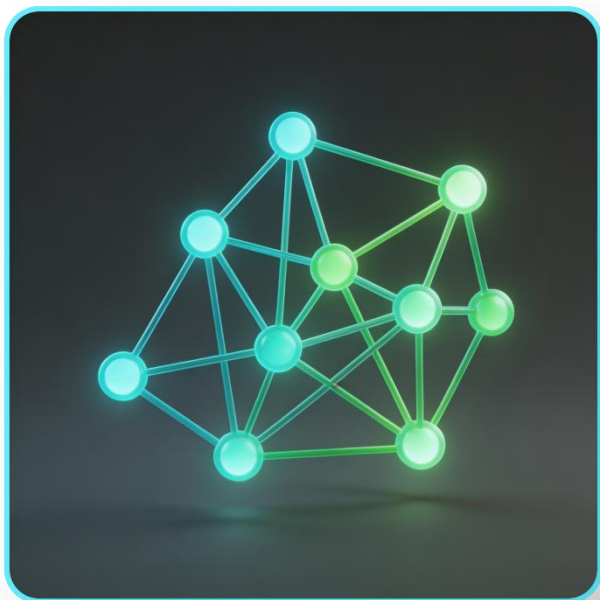
	HiF4	NVFP4
Storage Overhead	4.5 bits/value	4.5 bits/value
Group Size	64	16
Special Values	NaN and ± 0	NaN and ± 0
4-bit Element	S1P2 (E1M2)	E2M1
Significand Precision	3 bits	2 bits
Global Base Scale	E6M2	E4M3
Max Positive Value	$2^{18} \times 1.3125$	$2^{11} \times 1.3125$
Min Positive Value	2^{-50}	2^{-10}
Global Dynamic Range	$[-50, 18]$: 69 binades	$[-10, 11]$: 22 binades
Local Dynamic Range	4.81 binades	3.58 binades



03

Technical Implementation





Core Implementation

Native implementation of HiFloat8 operators based on torch.npu, fully compatible with the PyTorch tensor operation system.



Mixed-Precision Solution

Supports flexible mixed-precision training with "HiFloat8+FP8+FP16". Developers can choose formats as needed without refactoring the training process.



Ecosystem Adaptation

Adapted to mainstream open-source training frameworks, compatible with large model distributed training paradigms.





Solution Introduction

A flexible HiFloat8 simulation environment built on **torchao**. No specialized hardware is needed; performance and accuracy can be evaluated on existing mainstream GPUs.

This approach removes the hardware barrier, allowing researchers and engineers to seamlessly integrate and test HiFloat8 in their current workflows.



Core Capabilities

- Quickly simulate conversion and rounding from high-precision formats to HiFloat8.
- One-click comparison of training/inference effects between HiFloat8 and FP16/FP8.
- Full support for simulation validation of mainstream open-source models like LLaMA and Mistral.

<https://github.com/global-computing-consortium/HiFloat8>





MindSpeed Frameworks

The core of the MindSpeed low-precision training workflow lies in quantizing general matrix multiplication (GEMM) operations during forward propagation (Fprop), activation backward propagation (Dgrad), and weight backward propagation (Wgrad) to run at FP8 / HiF8 precision.

While the overall training process still operates on a BF16 / FP16 Automatic Mixed Precision (AMP) mode, specific computational operators—particularly the matrix multiplications within Linear layers (covering Fprop, Dgrad, and Wgrad)—are executed via an FP8 / HiF8 precision path.



TorchTitan Frameworks

TorchTitan, the PyTorch community's native distributed training framework for GPUs, offers several key advantages:

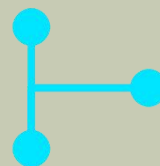
- **Flexible 3D Parallelism:** Enables modular, composable 3D parallelism with elastic scaling to handle varying workloads.
- **Built for Production:** Comes equipped with robust logging, checkpointing, and debugging tools ready for large-scale training tasks.
- **Optimized Performance:** Maximizes hardware efficiency through hardware-software co-design, including Float8 training and Symmetric Memory support.
- **Expert Engineering Insights:** Provides a proven blueprint for choosing and combining distributed training techniques based on real-world TorchTitan practices.





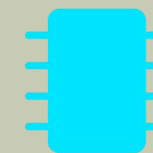
LLM Pre-training

HiFloat8 single-format full-process training for trillion-parameter models. Reuses FP16 delayed scaling strategy, reducing hyperparameter tuning costs by 80%, with stability matching FP16.



Open-Source LLM Fine-tuning

Fine-tuning mainstream 7B-13B open-source models with accuracy loss $< 0.1\%$. Memory usage is halved, allowing larger batch sizes on a single card.



Edge Agent Deployment

HiFloat8 quantized models achieve lossless inference on edge devices without complex calibration, suitable for offline, low-power edge agent scenarios.

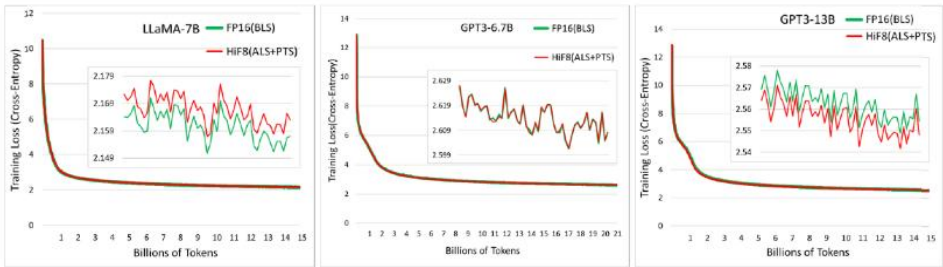


Compared with FP16, HiF8 performs well in multiple model training and inference tests

HiF8 Model Training

Convergence :

HiF8 training convergence and Loss curves are consistent with FP16



Accuracy:

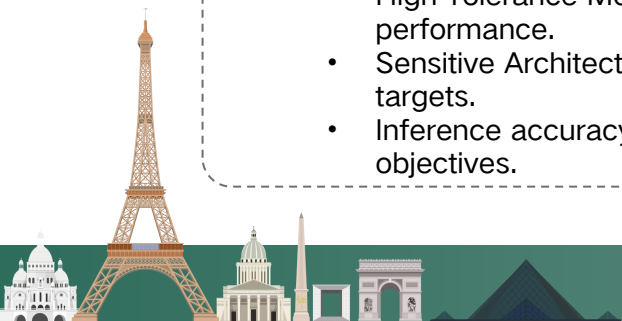
Training results: Minimal accuracy drop vs. 16-bit (difference within 0.2).

Table 7: Validation Perplexity for Large Language Models

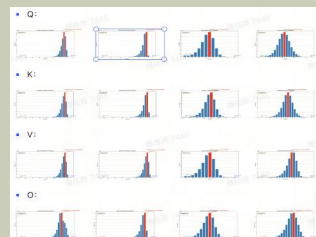
Model	Training Data	Training Tokens	HiF8 Strategy	FP16	HiF8	HiF8 - FP16
T5-11B	Wiki	3.28B	BLS	7.40	7.53	+0.13
GPT3-350M	Wiki, OpenWeb, Book3	14.8B	BLS	17.43	17.52	+0.09
GPT3-2.7B	Wiki, OpenWeb, Book3	10.5B	BLS	14.96	15.04	+0.08
GPT3-6.7B	Wiki, OpenWeb, Book3	21.3B	ALS	13.06	13.14	+0.08
GPT3-6.7B	Wiki, OpenWeb, Book3	21.3B	ALS + PTS	13.06	12.99	-0.07
GPT3-13B	Wiki, OpenWeb, Book3	14.8B	ALS + PTS	12.62	12.54	-0.08
LLaMA-7B	Wiki, OpenWeb	14.8B	ALS + PTS	8.25	8.28	+0.03
LLaMA-13B	Wiki, OpenWeb	8.85B	ALS + PTS	8.71	8.74	+0.03

HiF8 Model Inference

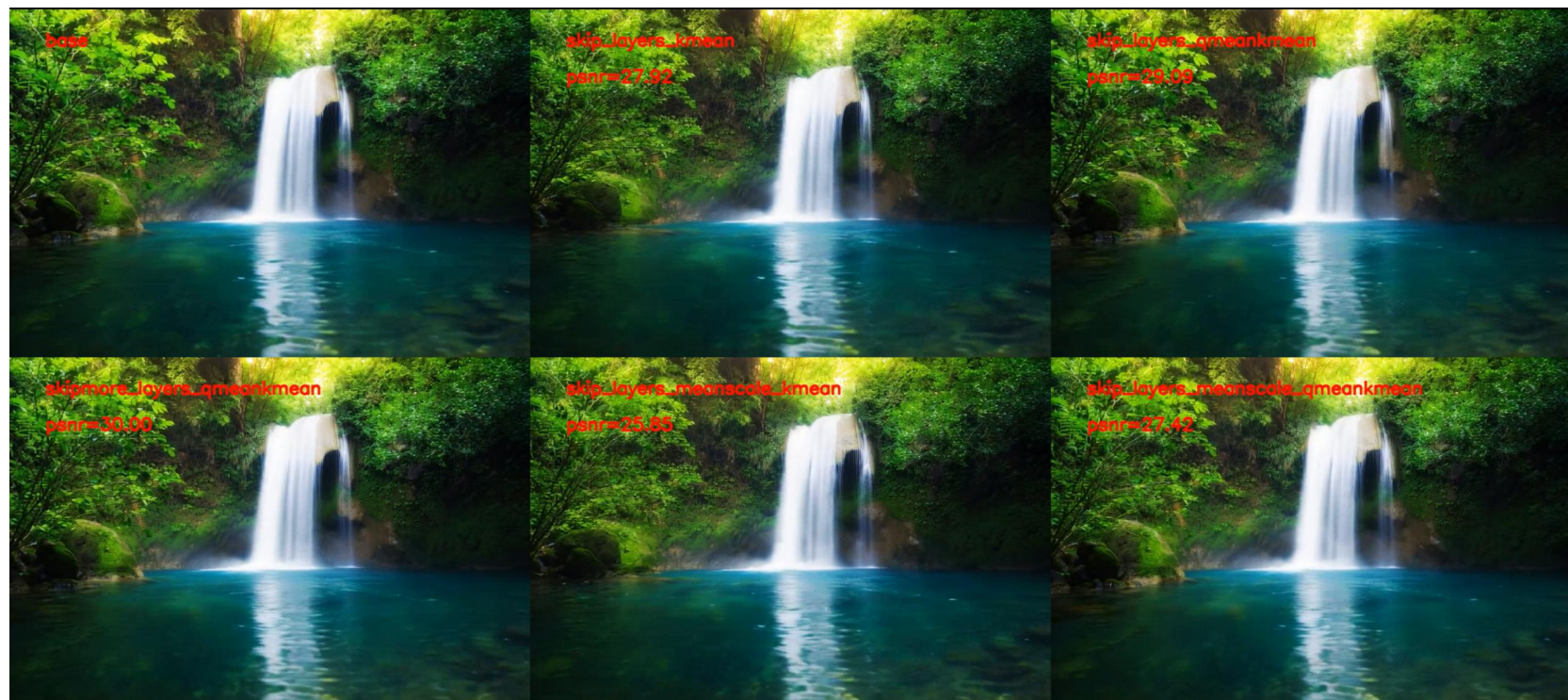
- High Tolerance Models: For quantization-resilient LLMs (e.g., Llama-65B), direct inference via HiF8 conversion yields optimal performance.
- Sensitive Architectures: For quantization-sensitive models like OPT, HiF8 paired with specific algorithms meets all predefined design targets.
- Inference accuracy: Assessment of downstream task accuracy on GPT/Llama reveals minimal precision degradation, aligning with our objectives.



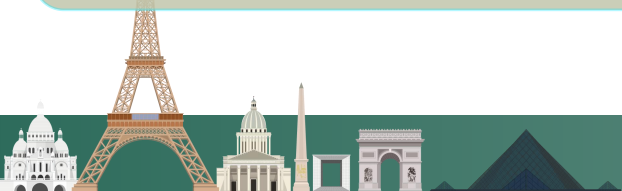
- **Data Analysis and HiF8 Quantization Scheme Design:**



- **HiF8 Quantization for Linear Layers (Per-tensor Current Scaling):** Based on data analysis, adopting per-tensor current scaling for linear layers meets the required training precision.
- **Delayed Scaling Training:** HiF8 quantization is implemented using Per-Tensor Scaling (PTS) across linear layers, optimizers, and collective communication modules. For Flash-Attention (FA) and Flash-Attention-Gradient (FAG), we employ a hybrid approach combining PTS and Per-Block scaling. Coupled with the Delayed Scaling algorithm, this ensures zero accuracy degradation.
- **Experimental Results:** Results confirm that 85% of layers support HiF8 direct-push without loss of accuracy. The outlier layers (remaining 15%) are managed via fine-grained scaling.



- (1) Original Image;
- (2) K + sageattention smooth(**skip 5% sensitive layer**);
- (3) QK + sageattention smooth(**skip 5% sensitive layer**);
- (4) QK + sageattention smooth(**skip 15% sensitive layer**);
- (5) QK + sageattention smooth(**95% hif8, 5% per-tensor scaling algorithm1**);
- (6) QK + sageattention smooth(**95% hif8, 5% per-tensor scaling algorithm2**).



04

Conclusion





Core Technology

The HiFloat8 tapered precision format fundamentally solves the core pain points of standard FP8 (instability, complexity), complementing rather than replacing the existing FP8 ecosystem.



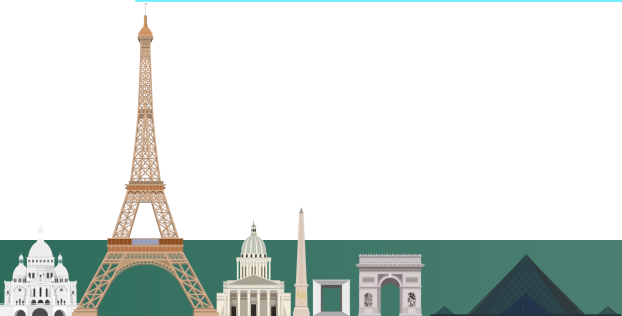
Developer-Friendly

Natively compatible with the PyTorch open-source ecosystem, enabling zero-threshold simulation and validation via torchao without sacrificing developer productivity.



Full-Scenario Deployment

Significantly improves the robustness of large-scale distributed system training for LLMs, while providing an efficient, low-cost quantization solution for edge agent deployment.

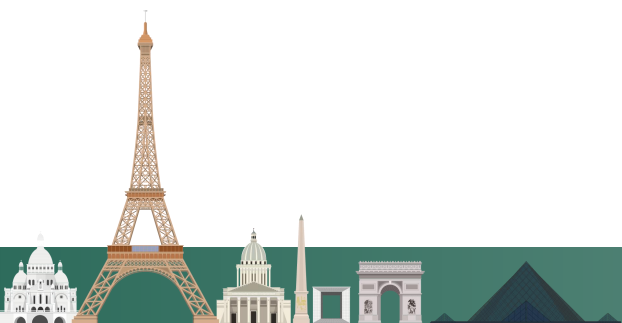


HiFloat Open-Source Project | Technical Exchange & Collaboration

Open-Source Repo: <https://github.com/global-computing-consortium/HiFloat8>
<https://github.com/global-computing-consortium/HiFloat4>

- Contact: zhaoyun67@huawei.com
- White Paper: arXiv:2409.16626v2
arXiv:arXiv:2602.11287v2(HiF4)

GOSIM Paris 2026 | Thank You



Thanks!

